

Task 1: Introduction to Cybersecurity

1.1. What is Cybersecurity?

- Cybersecurity is the practice of protecting computers, networks, devices, applications, and data from cyber threats.
- It prevents:
 - Unauthorized access
 - Data theft
 - Data damage
 - System disruption
- It uses security tools, techniques, and policies to protect digital systems.

1.2. Importance of Cybersecurity

- Protects sensitive and personal data.
- Prevents unauthorized access.
- Protects against cyberattacks.
- Prevents financial losses.
- Maintains user privacy.
- Ensures systems and services remain available.
- Protects an organization's reputation and business operations.

1.3. CIA Triad

The CIA Triad is a basic cybersecurity security model consisting of:

Confidentiality:

- Ensures information is accessible only to authorized users.
- Example: Password-protected files.

Integrity:

- Ensures data remains accurate and unchanged by unauthorized users.
- Example: Preventing unauthorized changes to a bank balance.

Availability:

- Ensures systems and data are available when required.
- Example: Keeping an online banking service accessible to users.

CIA = Confidentiality + Integrity + Availability

1.4. Personal vs Organizational Security Risks

a. Personal Security Risks:

Risks that affect individual users and their personal information.

- Phishing
- Weak passwords
- Identity theft
- Malware
- Online scams
- Data theft
- Unsecured Wi-Fi

Example: An attacker steals a person's email password through a phishing link.

b. Organizational Security Risks:

Risks that affect an organization's systems, data, and operations.

- Data breaches
- Ransomware
- Insider threats
- DDoS attacks
- Malware
- SQL injection

- Unauthorized access

Example: Ransomware encrypts a company's important files and disrupts its operations.

1.5. 10 Common Cyber Threats

1. Phishing

- A technique used to trick users into revealing sensitive information.
- Usually carried out through fake emails, messages, or websites.
- Example: Fake bank login page.

2. Malware

- Malicious software designed to harm or gain unauthorized access to systems.
- Examples: Virus, Worm, Trojan, Spyware.

3. Ransomware

- Malware that locks or encrypts data.
- Attackers demand payment to restore access.

4. Password Attacks

- Attempts to steal or guess passwords.
- Examples: Brute-force, Dictionary attacks, Credential stuffing.

5. Social Engineering

- Manipulating people into revealing confidential information or performing unsafe actions.
- Example: Pretending to be a bank employee and asking for account details.

6. DoS/DDoS Attack

- Attempts to make a website or service unavailable by overwhelming it with traffic.
- DoS: Attack from one source.
- DDoS: Attack from multiple sources.

7. Man-in-the-Middle (MITM) Attack

- An attacker secretly intercepts communication between two parties.
- Can result in data or credential theft.

8. SQL Injection

- Attackers insert malicious SQL commands into vulnerable applications.
- Can allow unauthorized access to databases.

9. Insider Threat

- A security threat caused by a person who has authorized access to an organization's systems.
- Can be intentional or accidental.

10. Zero-Day Attack

- An attack that exploits a previously unknown or unpatched software vulnerability.
- Can be difficult to detect and prevent before a fix is available.